

**BEFORE YOU READ — ANSWER FROM MEMORY**

1. Name **three** features that distinguish Crohn disease from ulcerative colitis on gross exam.
2. Which IBD has **transmural** inflammation and non-caseating granulomas?
3. What is **absent** in the affected bowel segment in Hirschsprung disease, and how is it diagnosed?

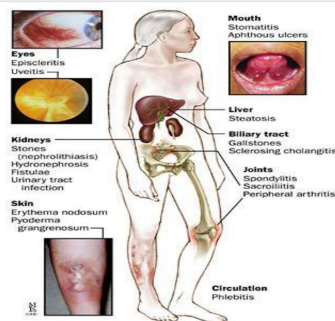
Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Inflammatory bowel disease (IBD) — the frame

What	Chronic conditions from inappropriate mucosal immune activation ; chronic, remitting-relapsing, of unknown etiology .
Two entities	Crohn disease (CD) and ulcerative colitis (UC) .
Pathogenesis (uncertain)	A combination of: altered host reaction to intestinal microbiota · intestinal epithelial dysfunction · aberrant mucosal immune responses · altered gut microbiome composition.
Clinical features	Altered bowel habits · rectal bleeding, urgency, tenesmus · abdominal cramping and pain · nausea and vomiting · systemic: weight loss, fever, sweating, malaise, arthralgia.

Extraintestinal manifestations

System	Manifestations
Eyes	Episcleritis · uveitis
Mouth	Stomatitis · aphthous ulcers
Liver	Steatosis
Biliary tract	Gallstones · sclerosing cholangitis
Joints	Spondylitis · sacroiliitis · peripheral arthritis
Kidneys	Stones (nephrolithiasis) · hydronephrosis · fistulae · UTI
Skin	Erythema nodosum · pyoderma gangrenosum
Circulation	Phlebitis



Extraintestinal manifestations, by system

2 Crohn disease vs ulcerative colitis — the discriminator grid

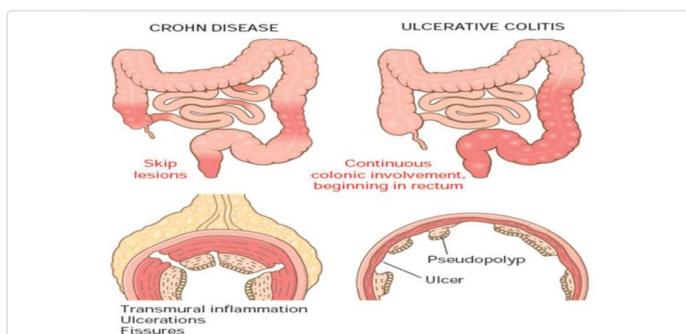
Distinction between CD and UC is based primarily on **morphology**.

Feature	Crohn disease	Ulcerative colitis
Macroscopic		
Bowel region	Any part, esophagus to anus (ileum ± colon typically)	Rectum and colon only
Distribution	Skip lesions	Diffuse, continuous
Stricture	Yes	Rare
Wall appearance	Thick	Thin
Microscopic		
Inflammation	Transmural	Limited to mucosa
Pseudopolyps	Moderate	Marked
Ulcers	Deep, knife-like	Superficial, broad-based
Lymphoid reaction	Marked	Moderate
Fibrosis	Marked	Mild to none
Serositis	Marked	Mild to none
Granulomas	Yes (~35%)	No
Fistulae / sinuses	Yes	No

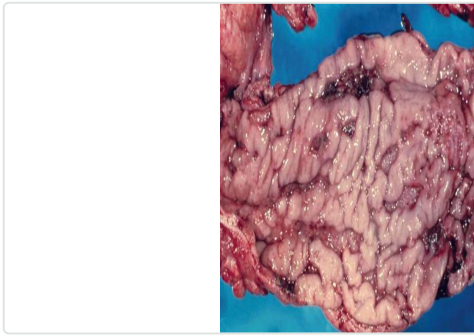
Clinical findings	CD	UC
Malabsorption	Yes	No
Toxic megacolon	No	Yes
Perianal fistula	Yes	No
Recurrence after surgery	Common	No
Malignant potential	Less	More
Smoking	Higher risk	Lower risk
Extraintestinal manifestations	Less	More

Feature	Crohn Disease	Ulcerative Colitis
Macroscopic		
Bowel region	Ileum ± colon	Colon only
Distribution	Skip lesions	Diffuse
Stricture	Yes	Rare
Wall appearance	Thick	Thin
Microscopic		
Inflammation	Transmural	Limited to mucosa
Pseudopolyps	Moderate	Marked
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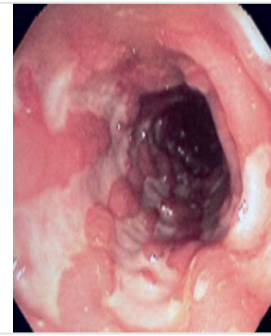
Macroscopic and microscopic discriminators (slide table)



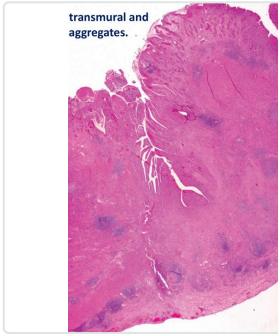
CD: skip lesions · UC: continuous, from rectum



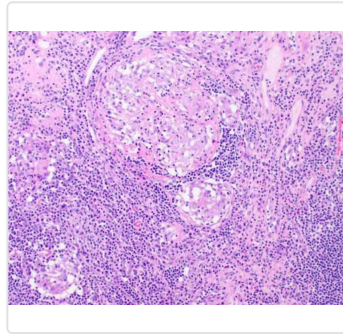
Crohn disease: cobblestone mucosa, longitudinal ulcers, intervening normal mucosa



UC: hyperemic mucosa; deep ulcer, endoscopic view



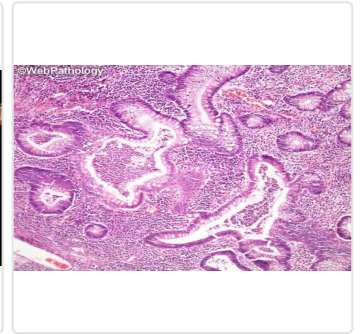
Crohn: transmural fibrosis, subserosal lymphoid aggregates



Crohn: non-necrotizing granuloma



Crohn: fissuring ulceration

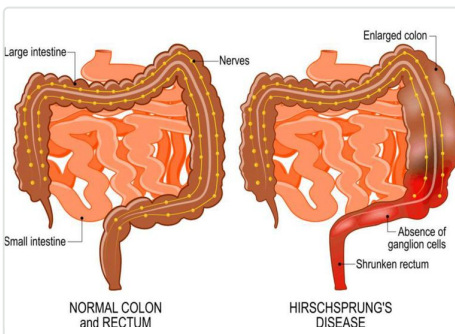


UC: crypt abscesses, crypt distortion, mucin depletion

UC colectomy: **continuous** involvement with an **abrupt transition** to normal mucosa; hyperemic mucosa with numerous **pseudopolyps**. Biopsy: **crypt abscesses**, marked inflammation, crypt distortion, regenerative atypia with mucin depletion.

3 Hirschsprung disease — illness script

Also known as	Congenital aganglionic megacolon ; occurs in about 1 in 5000 live births.
WHO gets it	Neonates: failure to pass meconium in the immediate postnatal period, which may progress to abdominal distention and bilious vomiting.
WHY it happens	Normal migration of neural crest cells from cecum to rectum is arrested prematurely , or ganglion cells undergo premature death → a distal segment lacking both the Meissner (submucosal) and Auerbach (myenteric) plexuses ("aganglionosis").
WHAT you see	Absence of peristalsis and functional obstruction . Contrast enema: affected segment is small caliber with proximal dilatation of the normal bowel. Rectal biopsy: absence of ganglion cells in the affected segment, with hypertrophied nerve fibers .
Complications	Enterocolitis · fluid and electrolyte disturbances · perforation and peritonitis.
Treatment	Surgical resection of the aganglionic segment, followed by anastomosis.



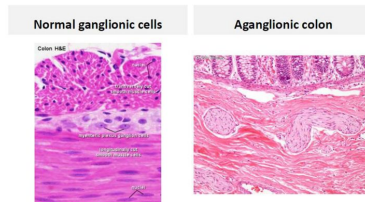
Normal colon vs Hirschsprung: absence of ganglion cells, shrunken rectum



Contrast enema: small-caliber segment with proximal dilatation



Neonatal abdominal distention



Diagnosis: rectal biopsy shows absence of ganglion cells within the affected segment with hypertrophied nerve fibers.

Rectal biopsy: normal ganglionic cells vs aganglionic colon (hypertrophied nerve fibers)

P Must-remember pearls

- **CD = skip lesions, transmural, granulomas, fistulae, ileum ± colon. UC = continuous from the rectum, mucosal only, crypt abscesses, no granulomas.**
- CD has **more malabsorption, strictures, perianal fistulae and surgical recurrence**; UC has **more toxic megacolon, malignant potential and extraintestinal manifestations.**
- **Smoking raises CD risk but is protective-associated with lower UC risk** (per the slide).
- **Hirschsprung = absent Meissner and Auerbach plexuses** from arrested neural crest migration → functional obstruction; diagnosis is by **rectal biopsy showing no ganglion cells.**
- **Contrast enema in Hirschsprung: the aganglionic segment looks narrow, with the normal bowel proximal to it dilated** — the opposite of what intuition suggests.

Q Test yourself

1. A colonoscopy shows skip lesions with deep, knife-like ulcers and a thickened bowel wall; biopsy shows transmural inflammation with a non-necrotizing granuloma. This is most consistent with:

- A. Crohn disease
- B. Ulcerative colitis
- C. Ischemic colitis
- D. Diverticulitis
- E. Hirschsprung disease

2. A colectomy specimen shows continuous mucosal involvement beginning in the rectum, with an abrupt transition to normal mucosa and numerous pseudopolyps. Biopsy shows crypt abscesses limited to the mucosa. This is:

- A. Crohn disease
- B. Diverticulitis
- C. Ischemic colitis
- D. Ulcerative colitis
- E. Hirschsprung disease

3. Compared with Crohn disease, ulcerative colitis carries a:

- A. Higher rate of perianal fistula
- B. Higher rate of recurrence after surgery
- C. Higher risk of malignant transformation
- D. Higher rate of malabsorption
- E. Lower risk of toxic megacolon

4. A newborn fails to pass meconium in the first two days of life and develops abdominal distention. Contrast enema shows a narrow-caliber rectosigmoid with proximal bowel dilatation. Which finding on rectal biopsy would confirm the diagnosis?

- A. Crypt abscesses with mucin depletion
- B. Non-necrotizing granulomas
- C. Transmural lymphoid aggregates
- D. Signet ring cells
- E. Absence of ganglion cells with hypertrophied nerve fibers

5. Hirschsprung disease results from arrested migration of which embryologic cell type?

- A. Endodermal stem cells
- B. Neural crest cells
- C. Mesothelial cells
- D. Hematopoietic stem cells
- E. Notochord cells

ANSWERS — COVER UNTIL YOU'VE COMMITTED

- 1 • A** — Skip lesions, transmural inflammation, deep ulcers, and granulomas are characteristic of Crohn disease.
- 2 • D** — Continuous, rectum-based, mucosa-limited disease with crypt abscesses and pseudopolyps is ulcerative colitis.
- 3 • C** — UC carries more malignant potential and more extraintestinal manifestations; CD has more fistulae, malabsorption, strictures and surgical recurrence, and is associated with a higher smoking risk.
- 4 • E** — Hirschsprung disease is confirmed by rectal biopsy showing absence of ganglion cells (aganglionosis) with hypertrophied nerve fibers.
- 5 • B** — Normal migration of neural crest cells from cecum to rectum is arrested prematurely, or ganglion cells die prematurely, producing aganglionosis.

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from LG3, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.